

MICROMOLD · DESIGN DECISION RECORD

MicroMold recommendations accepted

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DOCUMENT CONTROL

Document ID	MMD-DDR-002	Version	0.2
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/micromold		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget approved by Amish (\$520); O3 decided
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation remain "Proposed, awaiting Amish".

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." This record lists every MicroMold item that carried a recommendation in docs/REVIEW.md (all sessions) or in MMD-DDR-001, what changed in the repo because of it, and the items still open. Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, so any part of a decision that needs a build, a test, a measurement or purchasing is recorded here as decided but on hold. trl and trl_target stay at 3.

■ Options considered

The options for D1 to D10 are in MMD-DDR-001 and docs/REVIEW.md (session 2026-09-25, /populate). The options for N1 to N5 are in docs/REVIEW.md (session 2026-09-25, TRL 3, "Still awaiting Amish", items 3 to 7).

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Drive	Rack, pinion and head of a 1 t arbor press on a taller column; screw press kept as a variant. Includes the TRL 3 ratchet handle.	Wording only (MMD-DDR-001 v0.2, MMD-PRC-001 v0.4); design already in the model
D2	Layout	Vertical barrel, mold below	Wording only
D3	Injection	Plunger, not a reciprocating screw	Wording only
D4	Heating	Two PID zones, barrel and nozzle	Wording only
D5	Mold clamping	Bolted two-plate molds on a screw lift table; toggle clamp frame later. Includes the nozzle tip at 190 mm and the nozzle zone shield.	Wording only
D6	Pressure indication	10 kN load cell under the ram, on a G-11 spacer	Wording only
D7	First mold	64 x 50 x 6 mm test plaque; 4 mm nozzle and 5 to 7 mm sprue	Wording only
D8	Budget scope	Molds are tooling outside the machine budget (R14 covers the press)	Wording only; the figure is set by N1
D9	Fume control	Side hood with a duct fan, plus a written condition to run only under it or outdoors	Wording only
D10	Co-design partner type	A group that already shreds HDPE or PP	Wording only. Recruiting the partner is on hold with TRL 4
N1	Budget figure (R14)	Option (a): raise <code>budget_usd</code> to \$500 for the press, molds still as tooling	<code>project.yaml</code> <code>budget_usd</code> \$400 to \$500; README; R14 target \$400 to \$500 (MMD-REQ-001 v0.4)
N2	Mass target (R9)	Option (a): relax to 40 kg for a bench-bolted press	R9 mass target 35 kg to 40 kg (MMD-REQ-001 v0.4). Confirming the head mass of a real arbor press needs purchasing and a measurement: decided but on hold (TRL 4)
N3	Heater rating (R5)	Two 300 W barrel bands instead of 250 W	BOM item 7; MMD-PRC-001 v0.4; MMD-CAL-001 v0.2; media and drawing labels; heaters 600 W to 700 W
N4	Mold cooling (R6)	A small fan at the mold	New BOM item 18 (\$12.00), modeled in <code>cad/src/model.py</code> , on drawing MMD-DWG-001 Rev P2 and in the media; MMD-CAL-001 v0.2
N5	Feedstock grade (R1)	R1 requires flake from injection-molded items (caps, crates, buckets); bottle-grade HDPE excluded	R1 text (MMD-REQ-001 v0.4)

■ Effect on the numbers

Table 2. Before and after (MMD-CAL-001 v0.1 to v0.2).

Quantity	Before	After
<code>budget_usd</code>	\$400	\$500
Press parts cost (R14)	\$495, over by \$95	\$507, over by \$7
Press with one mold	\$585	\$597
Mass target and estimate (R9)	35 kg; 38.5 kg, not met	40 kg; 39.2 kg, at risk (98 % of the target, with an assumed 8 kg press head)
Warm-up (R5)	14.8 min, at risk	12.2 min, met on paper
Cycle and throughput (R6)	6.4 min, 9.4 per hour, mold about 92 °C, at risk	5.2 min (soak limited), 11.6 per hour, mold about 47 °C, met on paper
Connected load (R10)	635 W, 5.3 A at 120 V	753 W, 6.3 A at 120 V, still met
Energy per shot	About 21 Wh	About 20 Wh
Requirement counts	2 not met, 3 at risk, 6 met on paper, 4 met by design, 1 not verifiable	1 not met, 2 at risk, 8 met on paper, 4 met by design, 1 not verifiable

■ Consequences

- Controlled documents revised: MMD-PRC-001 v0.4, MMD-REQ-001 v0.4, MMD-CAL-001 v0.2 and MMD-DDR-001 v0.2. Drawing MMD-DWG-001 is at Rev P2. `bom/bom.csv` has 18 lines. No pitch or problem rewording was recommended, so the pitch and problem in `project.yaml` are unchanged.
- R14 was still not met, by \$7, because the mold cooling fan (N4) was added after the \$500 figure (N1) was recommended. Amish closed this on 2026-09-26 (see below).
- No decision needs another repo to change, so there are no cross-repo actions.
- Decided but on hold because TRL 4 is on hold: recruiting the co-design partner (D10), choosing and weighing a real arbor press (N2), and any build or trial of the heaters, fan or press.

■ Items still open

Table 3. Proposed, awaiting Amish.

#	Item	Status
01	The specific first co-design partner and its city or region. No recommendation was made.	Proposed, awaiting Amish
02	First product molds after the test plaque. For co-design; no recommendation was made.	Proposed, awaiting Amish
03	New: the press is \$507 against \$500. Options: (a) raise <code>budget_usd</code> to \$520; (b) keep \$500 and recheck against real quotations at TRL 4, since a \$7 overrun is within the accuracy of indicative prices; (c) count the mold cooling fan as workshop equipment outside the press budget, as the molds are. Recommendation: (b).	Decided by Amish, 2026-09-26: option (a), <code>budget_usd</code> \$520 (see below)

■ Budget approved, 2026-09-26

On 2026-09-26 Amish wrote, in chat: "i approve all the budget items."

- Budget set to \$520 to cover the priced BOM: decided by Amish, 2026-09-26. The priced press is \$507 (molds as tooling, D8), so R14 moves from not met to at risk, with a \$13 (2.5 %) margin on indicative prices. The press with one mold is \$597; the mold stays outside the budget.
- Files changed: `project.yaml` (`budget_usd` 500 to 520); MMD-REQ-001 v0.5 (R14 target and status); MMD-CAL-001 v0.3 and `docs/04-calcs/sizing.py` (cost section and status table; counts 0 not met, 3 at risk, 8 met on paper, 4 met by design, 1 not verifiable); MMD-PRB-001 v0.5 and MMD-PRC-001 v0.5 (budget figure); `README.md` ; `bom/bom-notes.md` ; `docs/REVIEW.md` .